



Impulse guard

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From Telegram Bot to Chrome Extension

Previous solution

- Telegram chatbot
- ML model predicted purchase impulsiveness
- User answered guided questions
- Feedback collected for retraining

Goal remained the same

Help users recognize impulsive purchases before buying.

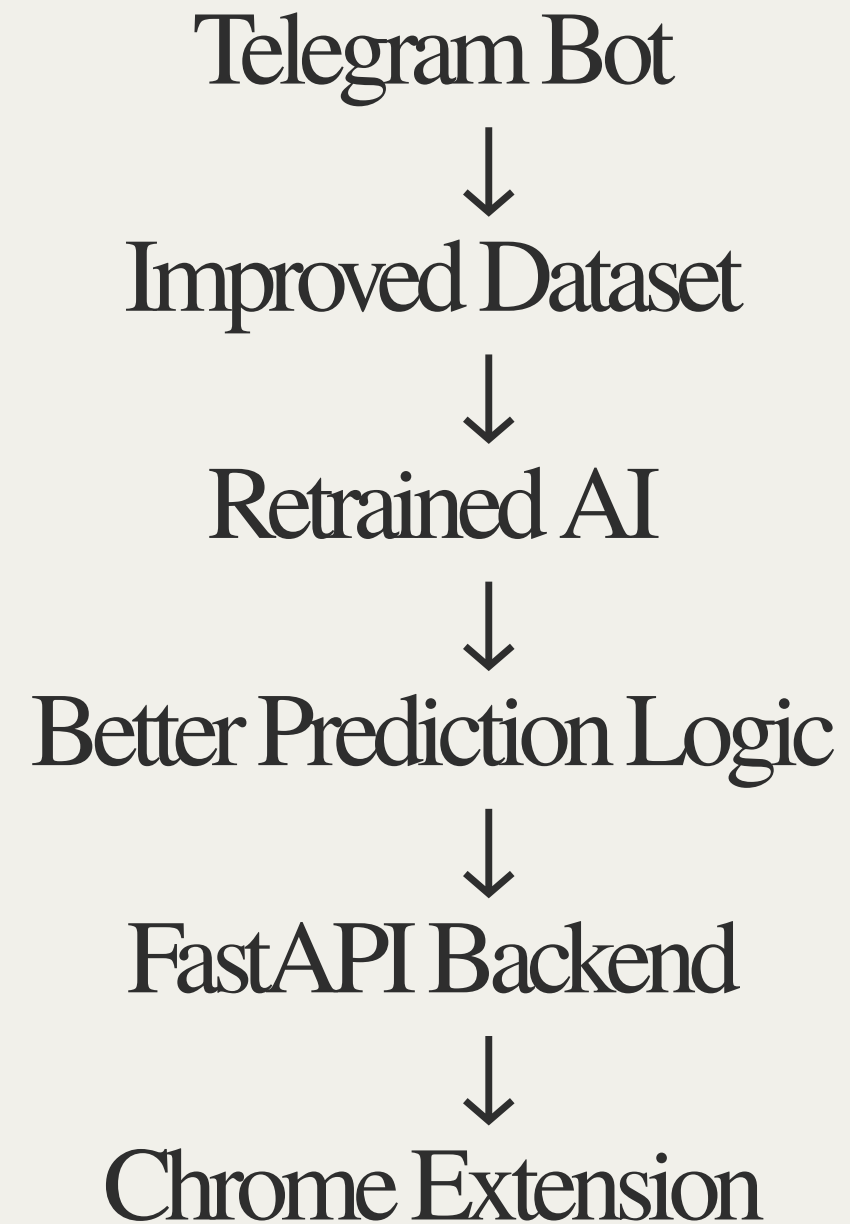
PROBLEMS WE IDENTIFIED



Challenge	Why it mattered
Limited integration with shopping websites	Users had to leave the shopping page to use the bot.
Conversation-based interaction	Too many steps before getting a prediction.
Prediction logic needed improvement	The AI could better distinguish planned purchases from impulsive ones.
Limited user experience	The interface was less intuitive for real shopping scenarios.
Professor's feedback	We were encouraged to create a more practical and realistic solution.

- These challenges motivated a complete redesign of both the AI system and the user interface, making ImpulseGuard more practical, accurate, and user-friendly.

HOW THE PROJECT EVOLVED



Major Improvements

- New Chrome extension
 - Retrained neural network
 - Improved prediction logic
 - New feedback system
 - Better UI
 - Automatic price detection
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New System Architecture



Why this architecture?

Modular and scalable

Easy to maintain

Real-time predictions

Reusable machine learning pipeline

Supports continuous improvement through user feedback

EVERYTHING IS CONNECTED THROUGH ONE MACHINE-LEARNING PIPELINE



Shopping Website



Chrome Extension



FastAPI Backend



Neural Network



Prediction



User Feedback



Future Retraining

AI IMPROVEMENTS:

Before:

- Simpler decision logic
- Limited interaction
- Older dataset

After :

- 10,000 synthetic purchases
- Retrained neural network
- Better preprocessing
- Improved impulse scoring
- Human feedback loop



Prediction now considers planning and necessity rather than only price

MODEL PERFORMANCE

Macro F1
 ≈ 0.67

Accuracy
 ≈ 0.68

Weighted F1
 ≈ 0.68

- MAJORITY BASELINE 0.12
- LOGISTIC REGRESSION 0.66
- NEURAL NETWORK 0.67



The model clearly outperforms the baseline while maintaining balanced performance across all classes.

What Influences the Prediction?



FEATURE IMPORTANCE:

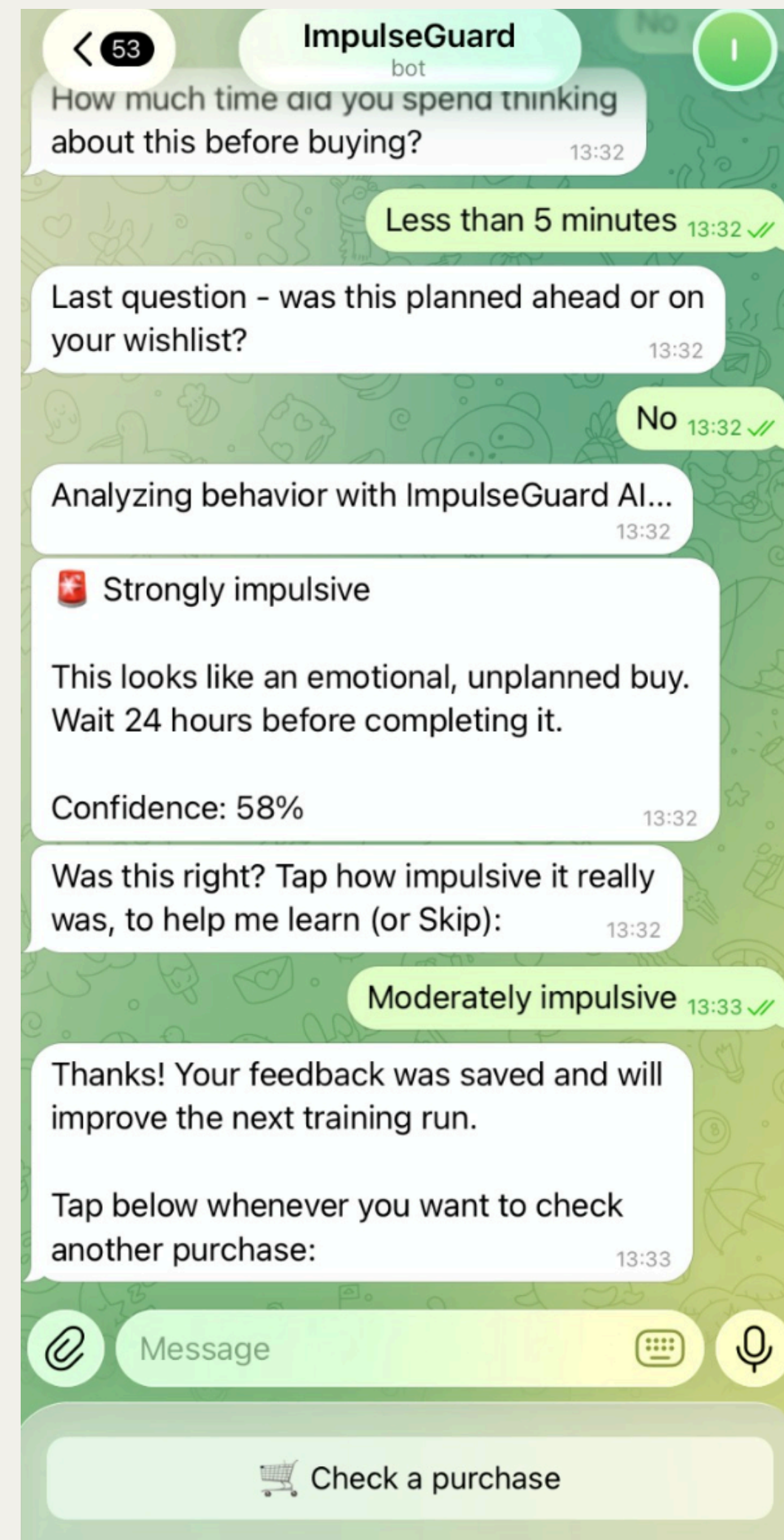


Planning and necessity influence impulsive buying much more than price.

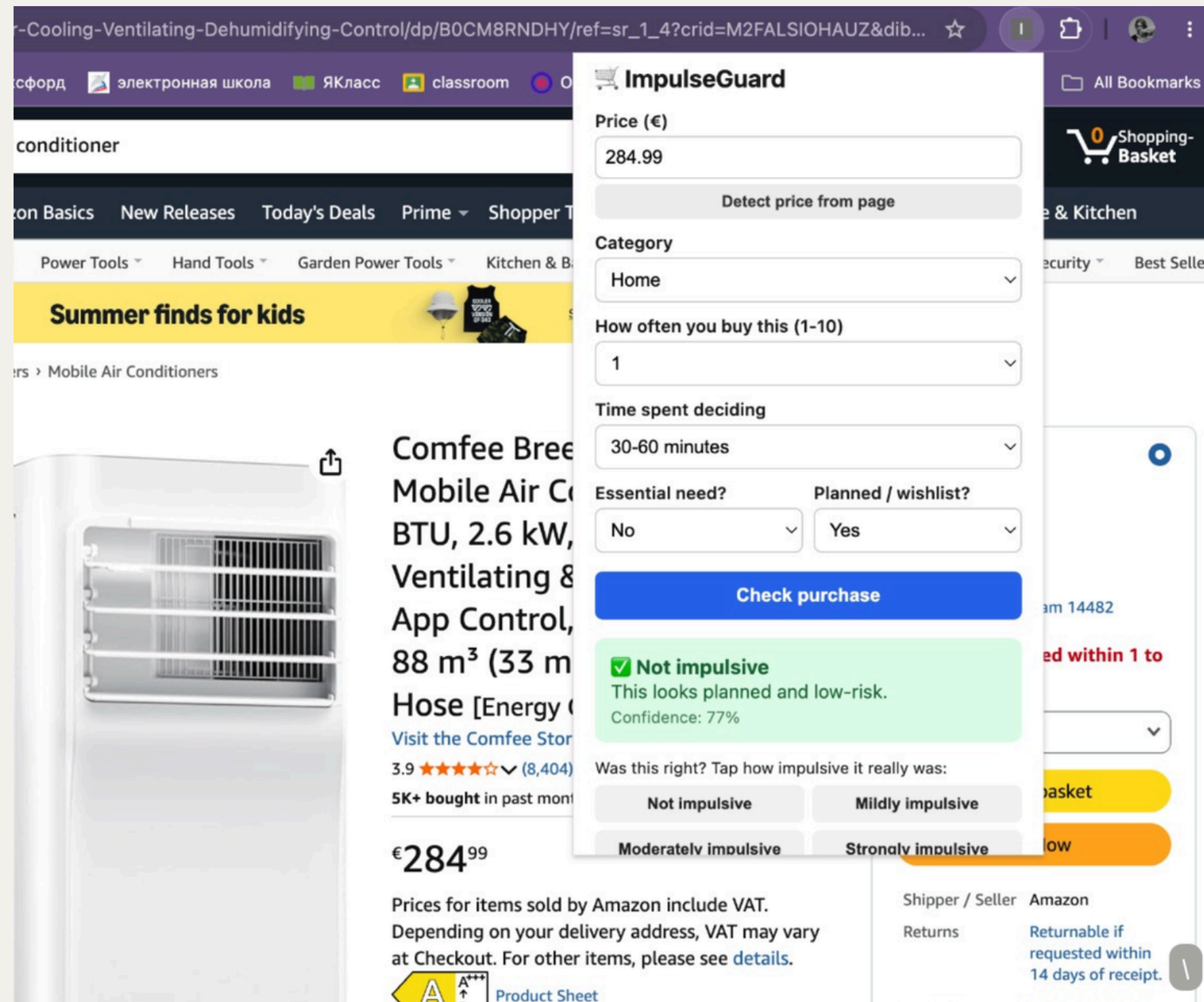
This is one of your strongest findings.

Telegram chatbot

- Mobile interaction
- Guided conversation
- Feedback collection

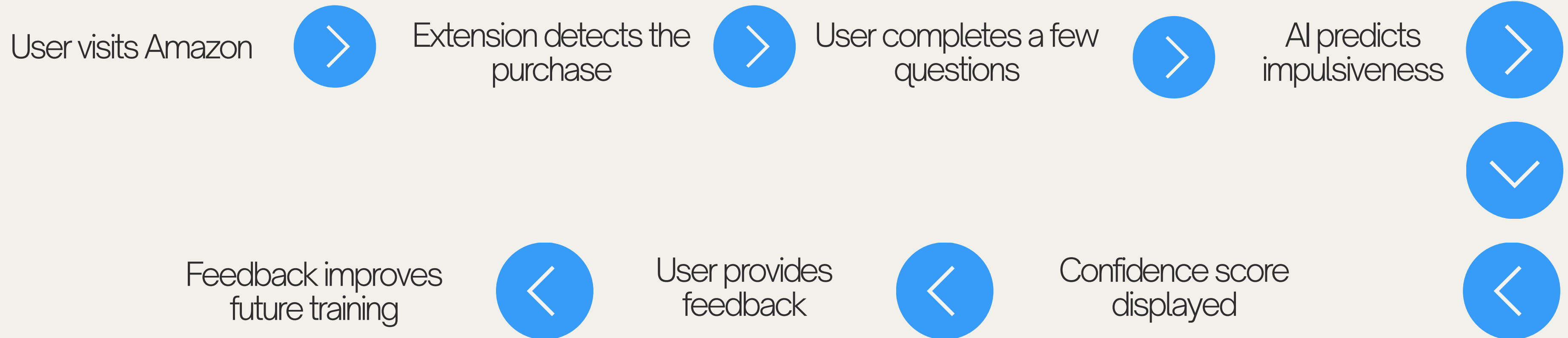


Chrome Extension



- Works directly while shopping
- Detects product price
- Faster decisions
- Better user experience

COMPLETE WORKFLOW



Final Outcome —

Smarter AI:

- Retrained neural network
- Chrome extension integrated with shopping websites
- User feedback supports future retraining
- Helps users make more conscious purchasing decisions

ImpulseGuard evolved from a simple chatbot into an intelligent browser assistant powered by machine learning.